

An End-to-End Dual-Attention Based Target Speaker Extraction Network for Explicit Optimization of Speaker Confusion

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Abstract—The task of target speaker extraction (TSE) involves isolating a specific speaker’s voice from a mixture of multiple audio sources. Most existing TSE methods primarily optimize the reconstruction quality of the extracted waveform while overlooking essential cues contained in the reference audio. Moreover, traditional TSE networks lack an interactive learning mechanism that jointly models mixed and reference embeddings, often leading to speaker confusion, particularly when there is a notable mismatch between the reference and target audio. To overcome these challenges, we propose a novel end-to-end dual-attention-based target speaker extraction (DATSE) network that explicitly addresses speaker confusion. DATSE employs an attention-weighted multiscale speech encoder that adaptively emphasizes features at different temporal resolutions. In addition, it introduces an attention-based speaker aggregation module within the auxiliary speaker recognition network, which effectively integrates speaker-specific information and leverages mixed-audio cues to refine target speaker embeddings. Experimental results on the WSJ0-2mix-extr dataset show that, the proposed method not only achieves competitive performance across multiple metrics but also significantly reduces speaker confusion under various disturbances such as noise and reverberation, thereby effectively lowering the overall confusion rate.

Impact Statement—Target speaker extraction (TSE) is essential for enhancing speech processing in real-world applications where background noise and multiple speakers pose challenges. This study introduces a dual-attention-based network (DATSE) that addresses speaker confusion and improves extraction quality in noisy and reverberant environments. Technologically, DATSE advances the field of speech processing by improving the robustness and accuracy of speaker extraction. Socially, it has the potential to enhance accessibility for individuals with hearing impairments

by isolating specific speakers in complex environments. Economically, it can reduce manual effort in fields like security or customer service, offering more efficient data processing. The proposed methods, therefore, advance AI knowledge in speech extraction and provide practical benefits in diverse, real-world scenarios.

Index Terms—Cross attention, multitask learning, speaker embedding, target speaker extraction (TSE).

I. INTRODUCTION

TARGET speaker extraction (TSE) refers to the task of isolating the voice of a specific speaker from a mixture of multiple speakers, regardless of whether the audio includes noise or reverberation [1]. This process is typically applied in environments where multiple speakers are present, with the goal of extracting the target speaker’s voice while suppressing background noise and interference. TSE plays a crucial role in various applications, such as speech recognition, meeting transcription, and voice analysis. For example, in the aftermath of a maritime accident, the analysis of voice recordings from the voyage data recorder (VDR) can be vital for investigating the cause of the incident [2]. The VDR captures conversations, communications, and operational commands between the captain, crew, and other individuals involved before and after the accident. By analyzing these voice recordings, investigators can gain valuable insights into the ship’s operations, decision-making processes, and communication leading up to the incident. However, challenges arise in the presence of environmental noise on the ship, such as mechanical noise, wind, sea waves, and overlapping voices from multiple speakers, all of which can obscure or interfere with the identification of the target speech.

Although neural network-based speech separation has made significant strides, it still requires knowledge of or estimation of the number of sources in the mixture [3]. In addition, it suffers from the problem of global permutation ambiguity, which refers to the arbitrary mapping between source speakers and output signals [4]. This ambiguity limits its practical applicability. In contrast, speaker extraction, a variant of speech separation, focuses on isolating a single target speaker’s voice at a time. By leveraging auxiliary cues to identify and extract only the target speaker’s voice, target speaker extraction naturally avoids the global permutation ambiguity problem. Moreover, it does

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not require prior knowledge of the number of sources in the mixture, making it a promising approach that has garnered increasing attention from researchers [5], [6].

Leveraging the powerful mapping capabilities of deep neural networks (DNNs) for speech processing, a wide range of target speaker extraction methods have emerged in recent years, including SpeakerBeam [7], SpEx [8], TD-SpeakerBeam [4], SpEx+ [9], WASE [10], SpEx++ [11], X-SepFormer [12], and X-TF-GridNet [13]. Despite recent advances in target speaker extraction, several key challenges remain. First, most existing systems rely on speaker-recognition-oriented embedding extractors, which mainly encode global speaker identity while discarding fine-grained temporal dynamics, phonetic cues, and prosodic patterns in the reference speech. This compression not only limits the ability of the extraction network to leverage detailed speaker characteristics, but also stems from the fact that speaker encoders are typically trained independently of the extraction process, without adaptive refinement using cues from the mixed speech. As a result, the learned speaker representations often lack contextual awareness, reducing robustness in acoustically complex or highly overlapping conditions [14].

Second, current embedding-based pipelines often depend on long or clean reference utterances to obtain stable speaker embeddings. This reliance is closely related to the use of fixed, single-scale temporal representations, which are insufficient to jointly capture short-term phonetic details and long-range acoustic patterns under noisy or casual recording conditions. Consequently, the requirement for high-quality enrollment speech increases the burden on users and restricts applicability in practical scenarios, where reference audio may be short, noisy, or captured opportunistically. Maintaining reliable performance under realistic enrollment conditions therefore remains a challenge.

Third, speaker confusion continues to be a critical issue, particularly when strong interference or speech overlap is present [15]. Although several studies have attempted mitigation [15], [16], [17], existing solutions—such as inference-time post-processing [12] or loss-function modifications—often treat speaker representation learning and speech extraction as loosely coupled stages. This separation introduces additional computational cost and becomes unreliable when both the mixed and reference signals contain interfering speakers. The lack of tight interaction between mixed-speech cues and speaker embedding generation fundamentally limits the ability to resolve speaker ambiguity, leaving confusion-resilient extraction an open research problem. While multiscale attention has been explored in prior speaker extraction methods such as SpEx [8], existing approaches limit attention to reweighting speech features across temporal scales. In contrast, the proposed DATSE framework adopts a dual-attention architecture, where attention is applied not only to multiscale speech representation learning but also to mixture-aware speaker embedding refinement. By enabling speaker embeddings to be adaptively updated using mixed-speech cues, DATSE establishes a tighter coupling between speaker modeling and extraction, which is critical for reducing speaker confusion.

Motivated by these observations, this work revisits the architectural design of target speaker extraction systems and advocates tighter coupling between reference speech encoding, mixed-speech analysis, and speaker representation learning. We propose a dual-attention framework that jointly addresses the above challenges by enabling adaptive, multiscale speaker modeling within an end-to-end extraction pipeline. Specifically, a scale-aware attention mechanism enhances speech encoding by dynamically weighting temporal features at different resolutions, while a cross-attention module integrates informative cues from the mixed speech to refine speaker embeddings. This unified design allows speaker representations to be both multiscale and mixture-aware, leading to improved robustness against speaker confusion and adverse acoustic conditions without introducing additional inference-time complexity.

To address the aforementioned issues, this article proposes an end-to-end dual-attention-based target speaker extraction network aimed at explicitly reducing speaker confusion. Unlike existing target speaker extraction pipelines that decouple speaker encoding and extraction, the proposed DATSE framework adopts a tightly coupled end-to-end design that enables continuous interaction between reference and mixture representations. The key contributions of this work are summarized as follows.

- 1) A tightly coupled end-to-end target speaker extraction framework is introduced, which enables continuous interaction between reference speech encoding, speaker representation learning, and speech extraction, addressing the limitations of loosely coupled pipelines.
- 2) A dual-attention mechanism is developed to enhance speaker awareness, in which scale-aware attention captures multiscale temporal speech features, and cross-attention leverages mixed-speech cues to adaptively refine speaker embeddings rather than keeping them fixed after reference encoding.
- 3) A gated temporal convolutional network is designed for speaker feature extraction, enabling effective modeling of both local correlations and long-range temporal dependencies, which helps suppress irrelevant temporal patterns and mitigate speaker confusion.

The remainder of this article is organized as follows. Section II provides an overview of related work. Section III presents the architecture of the proposed DATSE network. Section IV outlines the experimental setup and reports the results. Finally, Section V concludes the article.

II. RELATED WORKS

A. Network Structure of Target Speaker Extraction

Target speaker extraction aims to automatically isolate the target speaker's audio from a mixture of sounds, using registered audio of the target speaker as auxiliary information. The typical approach in mainstream research involves two key components: the extraction network and the auxiliary network. The auxiliary network (or speaker encoder) generates speaker embeddings by encoding the registered audio of the target speaker.

These embeddings are then used by the extraction network to isolate the target speaker's voice from the mixed audio.

A major focus of current research is enhancing the auxiliary network to more effectively learn and encode speaker embeddings. Approaches such as VoiceFilter [18] and the SpEx family [8], [9], [11], [19] have advanced the auxiliary network by upgrading from basic feedforward and recurrent neural networks to more complex convolutional neural networks (CNNs) with residual structures. These upgrades improve the ability of the auxiliary network to represent speaker embeddings more effectively. Another key area of research focuses on how the extraction network leverages these speaker embeddings to extract the target speaker's audio from the mixed input. Techniques like VoiceFilter, SpEx, SpEx+, and SpEx++ integrate speaker embeddings into the extraction process by adding them as biases to the semantic embeddings of the mixed audio. In contrast, SpeakerBeam [7] employs a multibranch network that combines speaker embeddings with the semantic embeddings of mixed audio in a weighted fashion.

However, existing methods face notable limitations. Most importantly, they lack a feedback mechanism in the extraction network that allows the semantic embeddings of the mixed audio to refine the speaker embeddings. Furthermore, while the auxiliary network relies solely on registered audio to learn speaker embeddings, it does not effectively incorporate mixed audio for auxiliary learning. This lack of interaction between the extraction and auxiliary networks can result in suboptimal performance, particularly when significant differences exist between the target audio in the mixed input and the registered audio, making it difficult for the auxiliary network to generate effective speaker embeddings based on registered audio alone.

B. Attention-Based Target Speaker Extraction Methods

Some researchers [20], [21] used an attention mechanism for dynamically computing the frame-wise speaker feature vector, a weighted summation of speaker embeddings from the enrolled target speaker. Kim [22] introduced Gaussian weight matrices into the multihead attention mechanism of the encoder to preserve both positive and negative correlation features in speech. Han [23] introduced an attention mechanism to the global bias vector of the target speaker and the embedding matrix of the mixed speech, aiming to leverage dynamic identity cues between different mixed audio and the target speaker. Hajavi [24] proposed a fine-grained early frequency attention mechanism for speaker representation learning, enabling the model to focus on significant regions of the input spectral representation to effectively capture speaker-specific information.

In recent years, some researchers [25], [26] have utilized the Transformer-based self-attention mechanism to extract contextual information. For example, Li [27] introduced an attention-based fusion method in the speaker extraction module, which adaptively integrates speaker embeddings. This method employs a self-attention mechanism, allowing the model to dynamically extract important features, thereby improving the quality of speaker embeddings. However, these models do not consider the interaction information between the registered audio and the

mixed audio. Although Wang [19] utilized an attention mechanism to selectively fuse features by considering the correlation between speaker embeddings and mixed audio, the speaker embeddings in the process do not participate in the updating.

III. PROBLEM FORMULATION AND FRAMEWORK OVERVIEW

A. Problem Formulation

TSE aims to recover the speech of a designated target speaker from a single-channel audio mixture that contains overlapping speech from multiple speakers as well as background noise. Let $m(t)$ denote the observed mixture signal of length T . This mixture consists of the target speech component $r(t)$ and L interfering sources $e_l(t)$, where each $e_l(t)$ represents either the speech of an interfering speaker or a background noise component. The mixture can be expressed as

$$m(t) = r(t) + \sum_{l=1}^L e_l(t), \quad t = 1, 2, \dots, T. \quad (1)$$

In addition, a reference utterance spoken by the target speaker is provided as auxiliary information. We denote this reference signal by $r(t)$ with length T' , i.e., $m(t) \in \mathbb{R}^T$ and $r(t) \in \mathbb{R}^{T'}$, where typically $T' \neq T$ and the linguistic content and acoustic conditions of the two signals may differ.

Given the mixture $m(t)$ and the reference speech $r(t)$, the objective of TSE is to estimate the clean target speech component $r(t)$ from the mixture. An end-to-end TSE system can thus be formulated as a mapping

$$\hat{r}_m(t) = \Phi(m(t), r(t)) \quad (2)$$

where $\hat{r}_m(t)$ denotes the extracted target signal and $\Phi(\cdot)$ represents the parametrized TSE model.

In practice, however, the reference utterance and the target speech embedded in the mixture may exhibit notable acoustic mismatch (e.g., in content, speaking style, or recording environment). When this mismatch is not properly modeled, the mapping $\Phi(\cdot)$ may produce an estimate $\hat{r}_m(t)$ that is closer to an interfering source than to the true target, leading to *speaker confusion*. Therefore, beyond accurate waveform reconstruction, the TSE problem considered in this work explicitly seeks a mapping $\Phi(\cdot)$ that is robust to reference–mixture mismatch and effectively suppresses speaker confusion while extracting the target speech.

The proposed DATSE framework, shown in Fig. 1, is composed of four major components.

- 1) *Attention-Weighted Multiscale Speech Encoder*: This module converts the time-domain mixture and reference signals into latent feature representations using multiple analysis window sizes. An attention mechanism adaptively weights different temporal scales, enabling the encoder to capture fine- and coarse-grained speech structures that are crucial for robust target speaker extraction.
- 2) *Attention-Based Speaker Encoder Network*: This network extracts speaker-discriminative embeddings by aggregating information from both the reference utterance

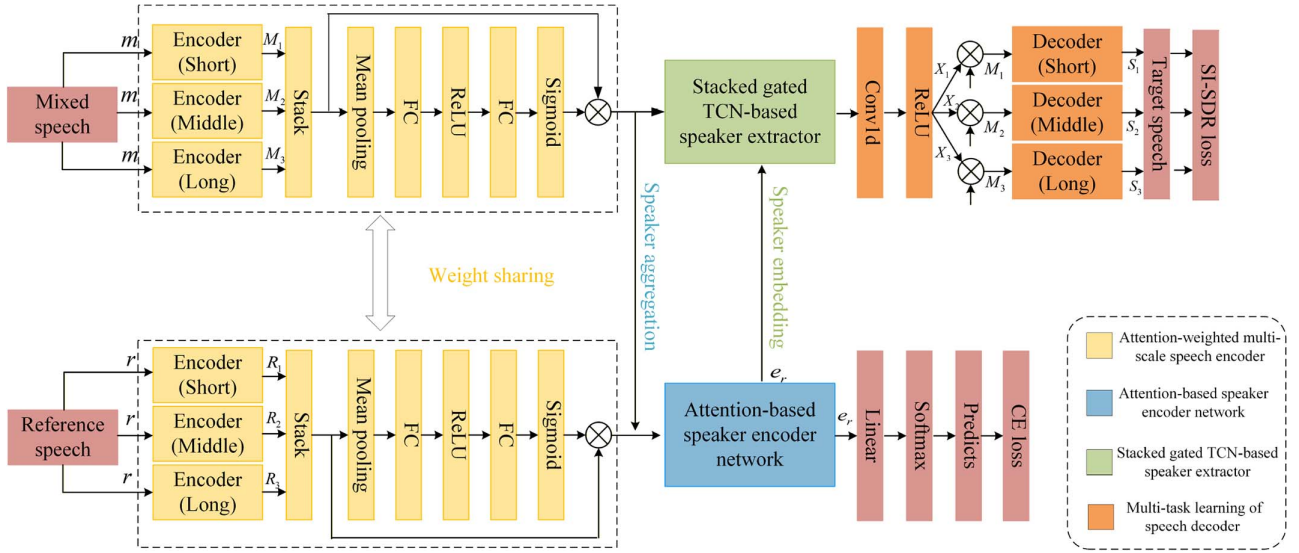


Fig. 1. Overall visualization of dual-attention based target speaker extraction framework. The four main components are represented by four different colors—yellow, black, green, and orange—and accompanied by captions. Due to space limitations, detailed steps for the black and green modules are shown in Figs. 2 and 3 below.

and the mixed audio. Through attention-driven aggregation and feature refinement, the network generates target-aware speaker embeddings that effectively guide the extraction process, even under acoustic mismatch between the reference and the mixture.

- 3) *Stacked Gated TCN-Based Speaker Extractor:* This component employs a stack of gated temporal convolutional network (gTCN) blocks to estimate speaker-dependent masks from the encoded features. The gating mechanism and dilated convolutions enable the extractor to efficiently model both local temporal patterns and long-range dependencies, producing accurate masks aligned with the target speaker characteristics.
- 4) *Multitask Learning of Speech Decoder:* This stage reconstructs the target speech signal from the estimated masks using a multiscale speech decoder, and jointly optimizes the entire model through a multitask objective. Specifically, the loss combines a scale-invariant speech reconstruction term with a speaker classification term, explicitly reducing speaker confusion while improving extraction quality.

The subsequent subsections elaborate on each component of the DATSE framework in detail.

B. Attention-Weighted Multiscale Encoder

Speech exhibits a rich temporal structure across multiple time scales, which captures the diversity of phonetics, prosody, and linguistic content [28]. Research has demonstrated that multitime-resolution speech analysis can significantly enhance the performance of speech recognition systems [19]. Motivated by this, and inspired by the multiscale time-domain speech encoder in SpEx+ [9], we employ three parallel 1-D convolutional layers, each operating at a different temporal resolution,

to encode both the mixed speech inputs $m(t) \in \mathbb{R}^T$ and the reference speech inputs $r(t) \in \mathbb{R}^{T'}$.

To encode the mixed and reference signals into the same feature space, we introduce the weight-shared multiscale speech encoders, also called twin multiscale speech encoders. Each of these convolutional layers has N filters with different kernel sizes: L_1 (short), L_2 (middle), and L_3 (long), respectively. By employing a weight-shared speech encoder to process both the mixed and reference audio, which are mapped to the same latent feature space, it becomes easier for subsequent speaker embedding extraction to fuse the features, thereby reducing the likelihood of feature mismatch. The rectified linear unit (ReLU) activation function is applied to these layers to produce nonnegative embedding coefficients as follows:

$$\begin{aligned} M_i &= \text{ReLU}(m * U_i), \quad i = 1, 2, 3, \\ R_i &= \text{ReLU}(r * U_i), \quad i = 1, 2, 3, \end{aligned} \quad (3)$$

where $*$ denotes the 1-D convolutional operator. $U_i \in \mathbb{R}^{N \times L_i}$ refers to the N convolutional filters that has a kernel size of L_i each. These filters are applied to the speech inputs at a fixed stride of $L_1/2$ samples to allow easy concatenation of the filter outputs at different temporal resolutions. $M_i \in \mathbb{R}^{N \times K_1}$ and $R_i \in \mathbb{R}^{N \times K_2}$ are the encoded embedding coefficients, wherein $K_1 = 2(T - L_1)/L_1 + 1$ and $K_2 = 2(T' - L_1)/L_1 + 1$ are the number of output frames. By performing multiscale analysis on the input audio, the model can utilize fine-grained features extracted from small analysis windows for more precise separation, as well as coarse-grained features captured from large analysis windows to stabilize speaker characteristics.

Furthermore, considering the varying importance of features at different scales in speaker recognition systems, we investigate attention-based multiscale weight learning. Specifically, we stack the three encoder features into $M \in \mathbb{R}^{3 \times N \times K_1}$ and

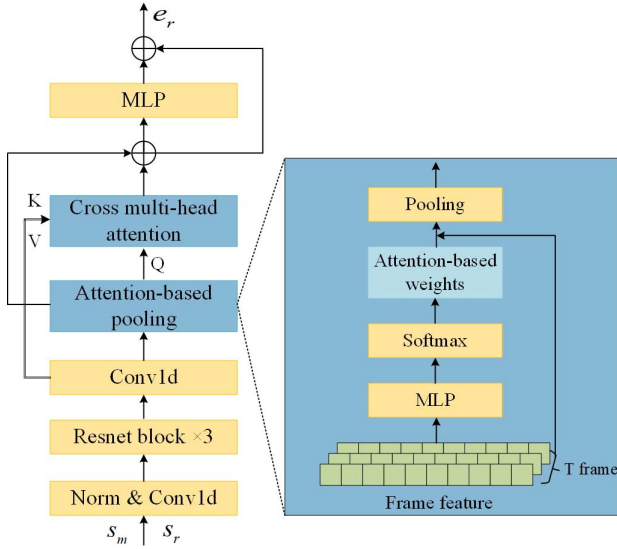


Fig. 2. Overview of the attention-based speaker encoder network.

pass it through a 2-D mean pooling layer for dimension reduction, followed by a squeeze-and-excitation (SE) block [29] to obtain the appropriate weight $W \in \mathbb{R}^{3 \times 1 \times 1}$ for each scale. This weight is used as a factor to calculate the weighted speech feature $M' \in \mathbb{R}^{3 \times N \times K_1}$ as follows:

$$M' = \langle W, M \rangle \quad (4)$$

where $\langle \cdot, \cdot \rangle$ means the dot product.

C. Attention-Based Speaker Encoder Network

After processing the mixed audio and registered audio with a multiscale encoder, we obtain the semantic embedding sequences, which will be input into an attention-based speaker aggregation auxiliary network to extract speaker embeddings, as shown in Fig. 2. First, the semantic embedding sequences are combined with the speaker frame-level extraction network based on the ResNet Block to obtain the frame-level speaker embedding sequences. Subsequently, the attention-based speaker aggregation module will sequentially aggregate certain features from the registered audio and target audio into the speaker embeddings. The specific steps are as follows:

Based on the attention mechanism of the shared multiscale speech encoder, we first construct a one-dimensional convolutional layer with a kernel size of 1×1 . Then, we apply normalization and feed the output into three identical residual blocks. Finally, we use a 1×1 one-dimensional convolutional layer and a pooling module to convert the frame-level feature representation into a fixed-dimensional speech-level representation.

It is important to note that when aggregating the frame-level speaker embedding sequences of the registered audio, we employ attention statistical pooling instead of the global average pooling used in SpEx+. The calculation process of attention statistical pooling is illustrated on the right side of Fig. 2. Attention pooling first utilizes the attention mechanism

to compute the weight for each frame. Subsequently, a weighted average is performed based on the frame weights, and the standard deviation of the weighted average results is calculated. The resulting speaker embedding is formed by concatenating the mean and standard deviation, resulting in a sequence of length 2. Compared with the global average pooling, attention statistical pooling takes into account the variations in speaker voice characteristics across different frames. This allows for effectively aggregating key frames containing the target speaker's voice features into the speaker embedding. In addition, the long-term variations of the speaker can be aggregated into the speaker embedding in the form of standard deviation. As a result, the speaker embedding obtained through attention statistical pooling has a more robust expressive capability and can effectively consolidate the key information in the registered audio.

Furthermore, although using attention pooling can enhance the recognizability of speaker embeddings from registered audio, there remains a possibility of bias in the speaker embeddings calculated solely from the registered audio when there is a certain disparity between the registered audio and the target audio. Moreover, because the target audio in mixed audio may be masked by interfering speakers, directly extracting speaker embeddings from mixed audio cannot obtain accurate sound characteristics corresponding to the target speech.

Therefore, this article employs a cross-attention mechanism, using the speaker embeddings extracted from registered audio as queries, while the frame-level speaker embeddings serve as keys and values input into the multihead attention model. The cross-attention mechanism calculates the correlation between the speaker embeddings and the semantic embeddings of each frame in the mixed audio, thereby identifying and perceiving the frames containing the target speaker within the mixed speech. As a result, the model can effectively aggregate certain characteristics of the target speaker from the mixed speech into the existing speaker embeddings without introducing excessive noise. Finally, by feeding the output of the multihead attention model into a multilayer perceptron, we can obtain speaker embeddings that are better suited for the target speech.

D. Stacked Gated TCN-Based Speaker Extractor

The deep dilated temporal convolutional networks (TCNs) network employs convolutional operations that allow for efficient parallel computation while providing the network with a very large receptive field by stacking the convolutional layers, allowing the network to capture temporal dependencies with a variety of resolutions of the input sequence [30]. Compared with RNN, gradient propagation in deep dilated TCNs network training is more stable, surpassing traditional LSTMs and GRUs across various tasks and datasets [31]. To simulate very long-term dependencies, the TCN architecture typically expands the receptive field, augmenting the count of hidden layers. Nevertheless, with a rise in the quantity of concealed layers, the ReLU activation function may experience the “dying” phenomenon in very deep architectures or under improper initialization conditions [32].

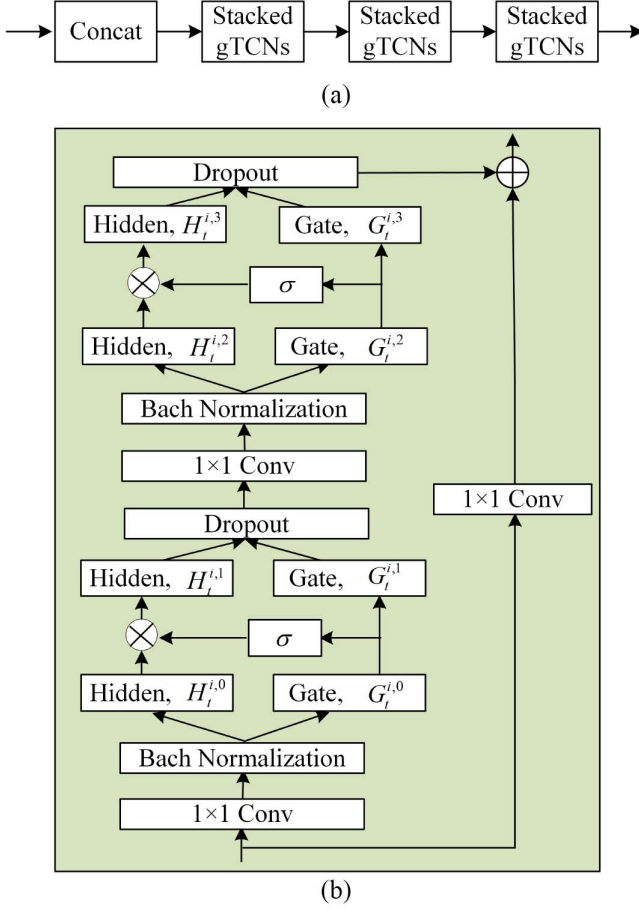


Fig. 3. (a) Details of the stacked gated TCN-based speaker extractor. (b) Details of the first gated TCN (gTCN) block in each stack. Each stack has 8 TCN blocks in this work. The remaining seven gTCNs don't take the speaker embedding as additional inputs.

Alsan [33] proposed a new gating mechanism for standard TCN architectures to model very long-term dependencies effectively. This approach modifies the gradient flow, addressing issues of vanishing or exploding gradients as well as the dead ReLU problem simultaneously. The architecture supports parallel training and generalizes well to variable-length input sequences and varying numbers of channels [34]. In this work, our speaker extractor repeats a stack of gated temporal convolutional network (gTCN) blocks with a number of 8 for 3 times. As shown in Fig. 3(b), the gated residual block on level j operates on the input sequence as follows:

$$[H_t^{j,0}; G_t^{j,0}] = \text{Conv1D}([H_t^{j-1,3}; G_t^{j-1,3}]) \quad (5)$$

$$[H_t^{j,1}; G_t^{j,1}] = [H_t^{j,0} \otimes \sigma(G_t^{j,0}); G_t^{j,0}] \quad (6)$$

$$[H_t^{j,2}; G_t^{j,2}] = \text{Conv1D}([H_t^{j,1}; G_t^{j,1}]) \quad (7)$$

$$[H_t^{j,3}; G_t^{j,3}] = [H_t^{j,2} \otimes \sigma(G_t^{j,2}); G_t^{j,2}] \quad (8)$$

where $\otimes$ is an operation for element-wise multiplication. We omit the weight normalizations, the dropouts, and the residual connection in the equations for notational simplicity.

The speaker extractor is designed to estimate masks X_i for the target speaker in each scale $i = 1, 2, 3$, conditioned on the embedding coefficients X_i and the speaker embedding e_r . Once

the masks K_i in various scales are estimated, the modulated responses $s_i = X_i \otimes M_i$ are obtained by element-wise multiplication of the masks X_i and the embedding coefficients M_i . We then reconstruct the modulated responses s_i into time domain signals y_i at multiple scales with the multiscale speech decoder as follows:

$$\begin{aligned} s_i &= d(X_i \otimes M_i) \\ &= d(f(M, e_r) \otimes M_i) \end{aligned} \quad (9)$$

where $f(\cdot)$ and $d(\cdot)$ are the speaker extractor to estimate the mask and the speech decoder to reconstruct the target signal, respectively.

E. Multitask Learning of Speech Decoder

The weight multiobjective loss is adopted to optimize the speaker extraction network via multitask learning

$$L(\Theta|m, r, s, I) = L_{SI_SDR} + \alpha L_{CE} \quad (10)$$

where Θ denotes the model parameters, m denotes the input mixture speech, r represents the reference speech of the target speaker, s represents the target clean speech, and I denote a one-hot vector representing the true class labels for the target speaker, and α is a scaling parameter.

L_{SI_SDR} aims to minimize the signal reconstruction error, which is defined as follows, with δ and γ as the weights to different scales

$$L_{SI_SDR} = -[(1 - \delta - \gamma)\rho(s_1, s) + \delta\rho(s_2, s) + \gamma\rho(s_3, s)] \quad (11)$$

$$\rho(\hat{s}, s) = 10 \log_{10} \frac{\|(\hat{s}^T s / s^T s) \cdot s\|}{\|(\hat{s}^T s / s^T s) \cdot s - \hat{s}\|} \quad (12)$$

where $\hat{s}$ and s denote the estimated signal and the target clean signal, respectively.

L_{CE} represents the cross-entropy loss for speaker classification, which is defined as

$$L_{CE} = - \sum_{i=1}^{N_s} I_j \log(\sigma(W \cdot e_r)_i) \quad (13)$$

where N_s denotes the number of speakers in the speaker classification task, I_i denotes the true class label for a speaker, and W is a weight matrix. $\sigma(\cdot)$ represents a softmax function and $\sigma(W \cdot e_r)$ denotes the predicted probability.

IV. EXPERIMENTS

A. Dataset

The experimental database used in this study is derived from the WSJ0-2mix-extr dataset (https://github.com/xuchenglin28/speaker_extraction), which is an adaptation of the widely-used WSJ0-2mix database [35]. The WSJ0-2mix-extr dataset, specifically designed for speaker extraction tasks, consists of two-speaker mixtures generated from the WSJ0 corpus at a sampling rate of 8 kHz. This dataset provides a challenging benchmark

TABLE I
OVERVIEW OF THE TRAIN, VALIDATION, AND TEST SETS

Subset of Data Sets	Sample Size (Utterance)	Speaker Size (People)	Length of Time (h)
Training set	20 000	101	30.4
Validation set	5000	101	7.7
Test set	3000	18	4.8

for speaker extraction and separation models by simulating real-world scenarios with overlapping speech. We follow the official WSJ0-2mix-extr dataset protocol and use the exact training, validation, and test splits provided in the public GitHub implementation. These splits are identical to those used by all baseline systems, and no modifications or custom partitions were applied.

The experimental dataset's composition and statistics are summarized in Table I. For both the training and testing phases, the first speaker in the mixed audio (Spk1) is consistently designated as the target speaker, while the second speaker (Spk2) serves as the interfering speaker. To facilitate model training and ensure robustness in speaker extraction, additional audio samples are drawn randomly from the target speaker's other utterances, excluding the target audio itself, to function as registration audio. This registration audio is crucial for establishing speaker identity and guiding the model's attention toward the target speaker in the mixed audio.

By employing this well-structured dataset, the study ensures a robust and systematic evaluation of the proposed methodologies under diverse and realistic audio mixing conditions.

B. Experimental Setting

In this article's experiments, all target speaker extraction systems were implemented using the PyTorch framework. Each system processes 4-s audio segments as input, ensuring consistency in training and testing. Most systems were initialized with a learning rate of 0.001 and trained for a maximum of 100 epochs on the training dataset. For fairness, the SpEx+ baseline [9] was trained using the learning rate of $1e^{-4}$ specified in its original implementation, while all other methods used $1e^{-3}$. To adaptively optimize the training process, the learning rate was halved whenever validation performance showed no improvement for two consecutive evaluations. Furthermore, training was stopped early if validation performance failed to improve for six consecutive evaluations, preventing unnecessary computation. The other parameters are set as $\delta = 0.15$, $\gamma = 0.2$, $\alpha = 0.15$. Experiments were conducted on an NVIDIA RTX 3090 GPU using PyTorch 1.13 and CUDA 11.7. For fair comparison, all baseline methods were trained or fine-tuned under the same experimental configuration. Each experiment was repeated using 5 different random seeds to ensure reproducibility, and the reported results represent the mean performance across runs, with standard deviations provided to reflect variability across different initializations.

C. Evaluation Metrics

To enable meaningful and fair comparisons, we adopt four objective evaluation metrics commonly used in TSE. These include: 1) signal-to-distortion ratio (SDR); 2) scale-invariant SDR (SI-SDR); 3) perceptual evaluation of speech quality (PESQ); and 4) speaker confusion rate (SCR).

1) *Signal-to-Distortion Ratio (SDR)*: SDR is a standard metric for quantifying the distortion level in reconstructed signals during speech enhancement or separation. It measures the ratio of the target clean signal energy to the distortion energy introduced by the system. Higher SDR values reflect better extraction quality and lower residual interference.

2) *Scale-Invariant Signal-to-Distortion Ratio (SI-SDR)*: SI-SDR improves upon SDR by removing the influence of global scale differences between the reference and estimated signals. This scale normalization ensures that the evaluation remains consistent regardless of gain variations. As shown in (12), SI-SDR is widely used in end-to-end TSE models because it better reflects true reconstruction fidelity.

3) *Perceptual Evaluation of Speech Quality (PESQ)*: PESQ estimates human-perceived speech quality and produces a MOS-like score ranging from approximately -0.5 to 4.5 . It captures perceptual naturalness and intelligibility, complementing SDR- and SI-SDR-based distortion measures and providing insight into subjective listening quality.

4) *Speaker Confusion Rate (SCR)*: SCR measures the percentage of test mixtures where the extracted signal is closer to an interfering speaker than to the target in speaker embedding space. This metric directly evaluates the system's robustness against speaker confusion, particularly in noisy, reverberant, or mismatched reference-mixture conditions.

By combining these complementary metrics, the evaluation captures distortion level, scale-invariant reconstruction quality, perceptual fidelity, and robustness against speaker identity confusion.

D. Reference Baselines

To benchmark our proposed method, we selected six state-of-the-art systems representing the latest advancements in monaural target speaker extraction. These systems have demonstrated competitive performance on similar datasets and provide a robust comparison framework.

- 1) *SpEx* [8]: The multiscale time-domain speaker extraction network (SpEx) is a pioneering deep learning model designed for speaker extraction tasks. By operating directly in the time domain and processing audio signals at multiple temporal scales, SpEx effectively captures both short-term and long-term dependencies, significantly improving extraction performance.
- 2) *SpEx+* [9]: SpEx+ builds upon the SpEx framework by transforming it into a fully time-domain operational structure. The model introduces a learning-based attention mechanism, enhancing its ability to selectively focus on the target speaker, thereby improving the robustness and accuracy of extraction in challenging acoustic scenarios.

- 3) *WASE* [10]: The weighted attention speaker extraction (WASE) system targets the extraction of individual speakers' voices from mixed signals, particularly in noisy environments. WASE employs a dynamic weighting mechanism that prioritizes audio segments critical for accurately isolating the target speaker, thus improving performance in real-world conditions.
- 4) *X-Sepformer* [12]: X-Sepformer is an end-to-end speaker extraction network that incorporates a transformer-based architecture to optimize the separation of speakers. The model explicitly addresses speaker confusion by leveraging attention mechanisms, making it highly effective in scenarios with significant speaker overlap.
- 5) *SEF-Net* [12]: The speaker embedding free target speaker extraction network (SEF-Net) eliminates reliance on speaker embeddings, using raw audio features instead. This approach enhances adaptability to diverse acoustic environments and speaker characteristics while maintaining high performance in extracting target voices from overlapping signals.
- 6) *X-TF-GridNet* [13]: X-TF-GridNet is a deep neural network architecture designed to model complex time-frequency (T-F) spectrograms for both single- and multichannel speech separation tasks. Focusing on the T-F domain, it provides a versatile framework for improving the separation quality across various audio processing scenarios.

These reference baselines serve as critical comparisons, highlighting the effectiveness of the proposed method against established systems in the field of target speaker extraction.

E. Ablation Experiments

We conducted an ablation study to demonstrate the contribution of each component in the proposed framework. This analysis involved the following competing methods.

- 1) *Without Gated Mechanism*: In this variation, the speaker extraction process utilizes only the stacked temporal convolutional network, omitting the gating mechanism. This setup assesses the role of the gating mechanism in enhancing the separation and extraction performance by controlling information flow dynamically.
- 2) *Without Scale-Attention Mechanism*: This configuration executes the same multiscale analysis for the dual speech encoder coding but does not include the scale-attention mechanism. The omission of scale attention evaluates its importance in emphasizing relevant scales for robust speaker extraction.
- 3) *Without Cross-Attention Mechanism*: In this setup, the speaker embedding module excludes the cross-attention mechanism, which typically aggregates features from the registered audio into the speaker embedding. This modification examines how the lack of cross-attention affects the model's ability to utilize reference audio to refine target speaker extraction.
- 4) *Without Classification Loss (w/o CLS)*: In this setting, the auxiliary speaker classification loss introduced in the

TABLE II
QUANTITATIVE COMPARISONS WITH ADVANCED BASELINES IN TERMS OF SDR, SI-SDR, PESQ, AND SCR ON THE WSJ0-2MIX-EXTR DATASET

Methods	SDR (dB)	SI-SDR (dB)	PESQ	SCR (%)
SpEx	16.3±0.21	15.8±0.24	2.84±0.05	5.2±0.6
SpEx+	17.2±0.19	16.9±0.20	2.96±0.04	4.7±0.5
WASE	17.6±0.18	17.1±0.17	3.02±0.04	4.1±0.4
X-Sepformer	19.5±0.16	18.9±0.15	3.10±0.03	3.6±0.3
SEF-Net	18.8±0.17	18.3±0.16	3.05±0.03	3.9±0.4
X-TF-GridNet	20.4±0.14	19.7±0.13	3.14±0.02	2.9±0.3
Ours	20.7±0.11*	19.9±0.10*	3.21±0.02*	1.8±0.3*

Note: *indicates statistically significant improvement over the strongest baseline (X-TF-GridNet) with $p < 0.05$.

multitask learning objective is removed, while all other components remain unchanged. This ablation evaluates the role of explicit identity supervision in mitigating speaker confusion.

- 5) *Shallow TCN Extractor (Small Receptive Field)*: To assess the importance of long-range temporal modeling, we replace the stacked gated TCN extractor with a shallow TCN that has a significantly reduced receptive field. This variant isolates the contribution of long-range dependencies by limiting the model's ability to aggregate information across extended temporal contexts. A performance drop in this setting indicates the necessity of a sufficiently large receptive field for effective target speaker extraction.
- 6) *Without Short-Scale Encoder Branch*: This variant removes the shortest time-scale branch in the multiscale speech encoder, which primarily captures fine-grained local acoustic structures. This ablation evaluates how much the model benefits from local temporal detail when forming discriminative speaker representations. A degradation in performance under this configuration highlights the importance of local feature modeling.

These ablation experiments provide insights into how each mechanism contributes to the overall performance of the proposed framework, highlighting their importance in achieving state-of-the-art results in target speaker extraction.

V. RESULTS AND DISCUSSION

A. Quality Assessment of Speech Extraction

As shown in Table II, the experimental results demonstrate that our method outperforms all baseline methods, including SpEx, SpEx+, WASE, X-Sepformer, SEF-Net, and X-TF-GridNet, on the WSJ0-2mix dataset. Specifically, our approach achieves the best results with a signal-to-distortion ratio (SDR) of 20.7 dB and a scale-invariant signal-to-distortion ratio (SI-SDR) of 19.9 dB.

Key observations include.

- 1) *Comparison With Early Methods*: Compared with SpEx and SpEx+ introduced in 2020, our method improves SDR by 4.4 dB and 3.5 dB, and SI-SDR by 4.1 dB and 3.0 dB, respectively, showcasing significant advancements.

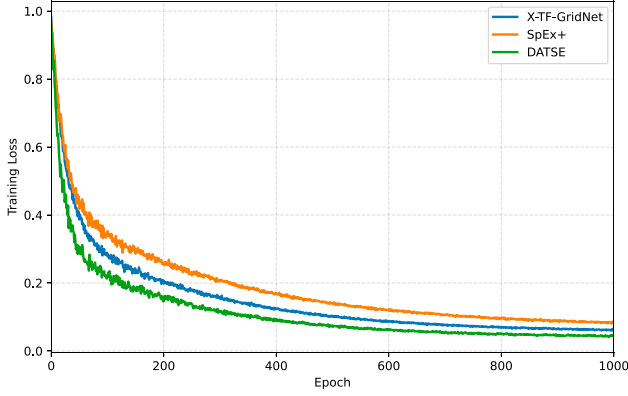


Fig. 4. Training stability and convergence analysis.

- 2) *Comparison With Recent Methods:* Among more advanced baselines, our method surpasses X-Sepformer (2023) by 1.2 dB in SDR and 1.0 dB in SI-SDR. Against SEF-Net (2023), the improvement is even more pronounced, with gains of 1.9 dB in SDR and 1.6 dB in SI-SDR.
- 3) *Comparison With the Latest Method:* Even compared with the state-of-the-art X-TF-GridNet (2024), our method shows marginal yet consistent improvements of 0.3 dB in SDR and 0.2 dB in SI-SDR.
- 4) *Statistical Significance Analysis:* Statistical significance analysis was conducted based on results from five independent runs using different random seeds. Paired-sample t-tests confirmed that the proposed DATSE achieves statistically significant improvements over the strongest baseline X-TF-GridNet across all evaluation metrics ($p < 0.05$). These results provide stronger statistical evidence supporting the reliability and robustness of the proposed framework.

Overall, these results highlight the robustness and generalization capability of our approach, establishing a new performance benchmark for target speaker extraction tasks.

To further verify that the observed performance improvements are not caused by experimental variability, we analyze the training dynamics of different methods. Fig. 4 shows the training loss curves of our method and several representative baselines averaged over multiple runs. As can be observed, the proposed method converges more rapidly and exhibits smoother optimization trajectories with lower variance across epochs. In contrast, baseline methods suffer from slower convergence and larger oscillations during training. This indicates that our model benefits from a more stable optimization process and improved generalization behavior. Combined with the consistent performance gains (reported as mean $\pm$ standard deviation over multiple runs), these results demonstrate that the improvements achieved by our approach are statistically reliable rather than the result of random fluctuations.

B. Results of Ablation Experiment

To analyze the contribution of each module in our model, we conducted ablation experiments by progressively removing specific components. The results, summarized in

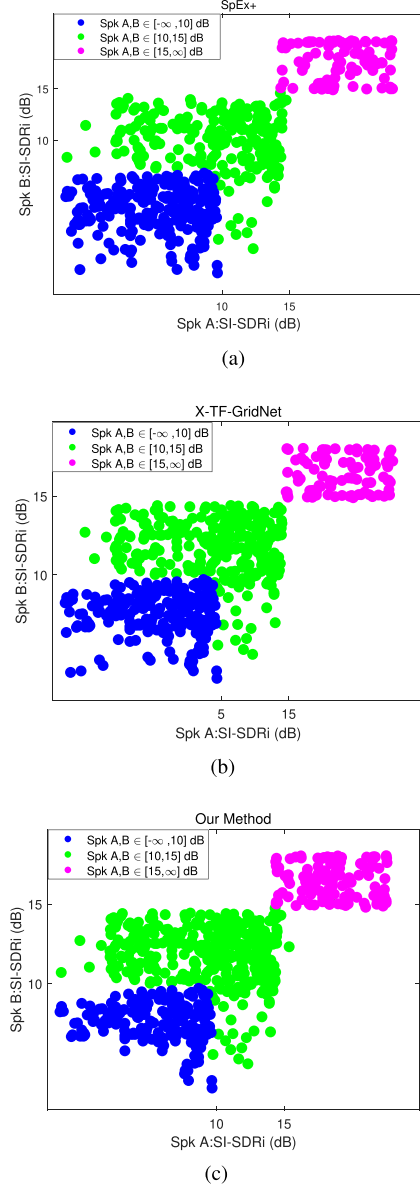


Fig. 5. Schematic diagram depicting SI-SDRi distributions among each mixture on the WSJ0-2mix dataset. That is, the joint distribution of the two speakers' performances on the SI-SDRi test set of Libri2 mixtures. In each audio mixture, the two speakers are set as targets in turn. A, B $\in [15, +\infty]$ dB (light black) indicated that both speakers achieved SI-SDRi; A, B $\in [-\infty, 10]$ dB (dark black) signified that both speakers scored below 10 dB. (a) SpEx+. (b) X-TF-GridNet. (c) Our method: DATSE.

Table III, demonstrate the impact of the Gated mechanism, scale-attention mechanism, and cross-attention mechanism on the overall performance.

Key findings are as follows.

- 1) *Without the Gated Mechanism:* Removing the gated mechanism results in a decrease in SDR to 19.5 dB and SI-SDR to 19.0 dB, highlighting its critical role in enhancing the model's separation capability.
- 2) *Without the Scale-Attention Mechanism:* The absence of the scale-attention mechanism causes a more pronounced performance drop, with SDR and SI-SDR decreasing to 19.0 dB and 18.5 dB, respectively. This indicates

TABLE III
ABLATION STUDY RESULTS IN TERMS OF SDR, SI-SDR, PESQ, AND SCR ON WSJ0-2MIX-EXTR

Methods	SDR (dB)	SI-SDR (dB)	PESQ	SCR (%)
Without gated mechanism	19.5±0.18	19.0±0.17	3.05±0.04	3.8±0.4
Without scale-attention mechanism	19.0±0.20	18.5±0.19	2.98±0.05	4.2±0.5
Without cross-attention mechanism	19.8±0.16	19.4±0.15	3.09±0.03	3.4±0.3
Without classification loss (w/o CLS)	19.6±0.17	19.1±0.16	3.07±0.04	3.1±0.3
Shallow TCN extractor (small RF)	19.1±0.19	18.6±0.18	2.99±0.05	4.0±0.4
Without short-scale encoder branch	19.3±0.18	18.8±0.17	3.01±0.04	3.7±0.4
W/o scale-attention + w/o gating	18.7±0.21	18.2±0.20	2.94±0.06	4.5±0.5
W/o speaker aggregation + w/o CLS	18.9±0.20	18.3±0.19	2.96±0.05	4.3±0.5
Complete model (Ours)	20.7±0.11	19.9±0.10	3.21±0.02	1.8±0.3

Note: Bold fonts indicate the best results.

the importance of scale-awareness in improving feature representation.

- 3) *Without the Cross-Attention Mechanism*: Excluding the cross-attention mechanism results in an SDR of 19.8 dB and an SI-SDR of 19.4 dB, showing that cross-attention contributes significantly but to a slightly lesser extent than other components.
- 4) *Without the Classification Loss (w/o CLS)*: Removing the speaker classification loss in the multitask learning module increases the speaker confusion errors and reduces performance to 19.6 dB SDR and 19.1 dB SI-SDR. This confirms that the classification objective effectively constrains the embedding space and plays a key role in mitigating speaker confusion.
- 5) *Shallow TCN Extractor (Small Receptive Field)*: Replacing the stacked gated TCN with a shallow TCN that has a greatly reduced receptive field leads to an SDR of 19.1 dB and an SI-SDR of 18.6 dB. This demonstrates the importance of long-range temporal modeling and verifies that a sufficiently large receptive field is essential for capturing extended contextual cues.
- 6) *Without the Short-Scale Encoder Branch*: Removing the short-scale branch from the multiscale encoder reduces performance to 19.3 dB SDR and 18.8 dB SI-SDR. This indicates that fine-grained local temporal structures also play a meaningful role in forming robust speaker representations.
- 7) *Without Scale-Attention + Without Gating (Two-Module Removal)*: Removing both the scale-attention mechanism and the gated TCN significantly degrades performance to 18.7 dB SDR and 18.2 dB SI-SDR, demonstrating that these two modules provide complementary benefits for capturing multiscale temporal cues.
- 8) *Without Speaker Aggregation + Without Classification Loss*: Disabling both the attention-based speaker aggregation and the classification objective increases speaker confusion and reduces performance to 18.9

dB SDR and 18.3 dB SI-SDR, highlighting the combined importance of embedding refinement and identity supervision.

- 9) *Complete Model*: Our full model achieves the best performance with an SDR of 20.7 dB and an SI-SDR of 19.9 dB, demonstrating the synergistic effect of all components working together.

These ablation results provide clear evidence of the functional contributions of each component in the proposed DATSE framework. First, removing either the gated mechanism or the scale-attention mechanism leads to a notable degradation in SDR and SI-SDR, illustrating their complementary roles in capturing multiscale temporal structures. The shallow TCN setting further confirms the importance of long-range temporal modeling, as the significantly reduced receptive field limits the model's ability to track extended contextual dependencies, resulting in a marked drop in performance. Likewise, removing the short-scale encoder branch weakens the representation of local temporal details, demonstrating that both fine-grained and long-range cues are essential for robust speaker extraction.

Importantly, eliminating the classification loss increases speaker confusion and causes an additional decline in both speech quality and extraction accuracy. This verifies the reviewer's insight that explicit identity supervision is crucial: the classification objective helps regulate the embedding space and prevents the extractor from drifting toward interfering speakers. The combined removal of key modules—such as jointly disabling scale-attention and gating, or disabling both speaker aggregation and classification loss—produces even larger performance drops. These multimodule ablations highlight the synergistic effects across components and confirm that each module addresses a distinct failure mode (local modeling, long-range modeling, scale-awareness, or speaker-identification consistency).

Overall, the full DATSE model achieves the highest performance across all metrics, demonstrating that the coordinated integration of multiscale encoding, attention-guided speaker representation, gated long-range extraction, and multitask learning

is essential for maximizing extraction quality while minimizing speaker confusion.

C. Visualization

To quantitatively validate the discussions above, we calculated SI-SDRi distributions for each mixture on the WSJ0-2mix dataset. Fig. 5 illustrates a schematic diagram depicting collective SI-SDRi distributions across the mixture.

From Fig. 5, it can be observed that the sample distributions predicted by SpEx+ and X-TF-GridNet are relatively dispersed. This indicates that DATSE exhibits more stable performance in extracting the target speaker compared with other existing models. In addition, as shown in Fig. 5, our proposed method significantly outperforms both SpEx+ and X-TF-GridNet in terms of the final SI-SDRi scores. This suggests that when these models use registered audio to extract the target speech, they not only fail to extract it effectively but also degrade the signal-to-noise ratio of the target speech relative to the interfering speech. As a result, our method demonstrates a clear advantage in terms of the final SI-SDRi score.

VI. CONCLUSION

This article proposes a speaker-aware dual-attention target speaker extraction network (DATSE). The core of the proposed method lies in the introduction of attention learning mechanisms in both the speech encoding and speaker embedding networks. In the speech encoding module, an attention mechanism is applied after a multiscale speech encoder to further extract effective features. In the speaker embedding module, a speaker aggregation module based on attention is designed to optimize the speaker embedding learning process by utilizing the acoustic characteristics of the target speech in mixed audio. In the speaker prior module, a stacked temporal convolutional module based on a gating mechanism is proposed to enhance the speaker-awareness capability. In the experiments presented in this article, the speech extraction quality of DATSE shows a significant improvement compared with the baseline models. In future research, further exploration will be conducted on better methods for time-frequency feature fusion, as well as addressing the issue of discrepancies between the output audio and the original audio in multispeaker mixed audio scenarios.

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